Xi Chen

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Research Profile

Statistics PhD candidate developing statistical machine-learning methods for graph-structured and time-series data, with applications in climate and physiological monitoring.

Education

University of California, Santa Cruz	PhD, Statistics; candidacy 07/2026	2023–Expected 06/2028
University of California, Davis	MS, Biostatistics	2021–2023
Sun Yat-Sen University	BS, Applied Mathematics	2014–2018

Technical Expertise

Methods: graph-fused LASSO, ADMM, AR/VAR models, weighted U-statistics, MCMC, Langevin dynamics, deep learning (CNNs, LSTMs/RNNs, neural decoders, Transformers)
Computing: Python (NumPy, pandas, PyTorch, scikit-learn), CUDA, R (Rcpp), SQL, MATLAB, Git, Linux, SSH

Research Experience

PhD Candidate & Graduate Researcher Sep 2023–Present
University of California, Santa Cruz

Latent-Space Graph-Fused LASSO *Preprint under revision*

- Co-developed an unsupervised method to cluster nodes using both graph connections and node attributes; a shared neural decoder models attributes, while a graph-fused penalty encourages connected nodes to share latent cluster structure.
- Implemented ADMM optimization and Langevin latent-variable inference. On a 144-node simulation, the method reached 97.0% normalized mutual information versus 83.1% for the strongest listed baseline.
- Applied the method to 58 California counties using shared borders and 168 monthly temperature observations, identifying 10 climate regions. arXiv:2511.04859.

GraphClustAR: Nodal Clustering in Weighted Graphs with Time Series *Manuscript in preparation*

- Co-developing an interpretable method to cluster graph nodes by their time-series dynamics: node-specific AR(p) models capture temporal patterns, and a weighted graph-fused penalty groups nodes with similar dynamics.
- Deriving finite-sample error bounds and implementing ADMM estimation in R; using BIC to select the penalty and a silhouette score to select the number of clusters.

Multivariate Changepoint Detection via Weighted U-Statistics *Manuscript nearing submission*

- Co-developing tests that identify when the means of several correlated time series change together, even when each variable shifts by a different amount or direction; applied to cardio-respiratory data from a CO2 inhalation experiment.
- Constructed weighted vector U-statistics to detect changes near series boundaries and studied their asymptotic behavior. Used VAR pre-whitening to address serial dependence and obtain dimension-based critical values.
- Compared CUSUM, Wilcoxon, and Huber tests in VAR simulations with up to 300,000 replications per setting; Rcpp made adjusted simulations about 4x faster than an R reference in a 1,000-run benchmark.

Research Assistant Sep 2019–Jul 2021
Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences

- **Blood-pressure estimation:** Trained supervised ECG/PPG models for potential wearable monitoring; filtered signals, segmented and aligned waves, and removed artifacts before training.
- **Arrhythmia detection:** Developed PPG-based deep-learning workflows using ECG-informed labels; a co-authored *JAHA* study classified six rhythms with 85.0% accuracy and 0.978 micro-averaged AUROC on 23,384 held-out waveforms.
- **Cardiac analysis:** Used linear mixed-effects models for repeated post-COVID-19 measurements, accounting for within-patient correlation.

Selected Publications & Preprints

- Liu, Z., Zhou, B., Jiang, Z., **Chen, X.**, et al. (2022). Multiclass arrhythmia detection and classification from photoplethysmography signals using a deep convolutional neural network. *Journal of the American Heart Association*, 11(7).
- Kei, Y. L., Padilla, O. H. M., Killick, R., Wilson, J., **Chen, X.**, and Lund, R. (2025). Decoder-only clustering in attributed graphs. arXiv:2511.04859. Preprint under revision.